Divyajeet Panda

Software Engineer | Backend Development | Data Systems

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience in backend services, data systems, and warehouse automation. Hands-on with Neo4j/Cypher, PostgreSQL, Java/Spring Boot, Python, and observability tooling. Current work centers on configuring facility-specific Cypher and SQL queries that translate CV scan payloads into client dashboard data, debugging data pipelines, validating UI outputs, and automating warehouse workflows. Prior enterprise experience includes microservices, REST and GraphQL APIs, event-driven systems, and cloud deployments across AWS and Azure.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, Cypher

Backend: Java/Spring Boot, FastAPI, Node.js, REST APIs, GraphQL APIs

Frontend: React, Next.js

Databases: Neo4j, PostgreSQL, MySQL, Graph Data Modeling, Graph Data Science

AI & LLMs: AI Agents (Microsoft Foundry, Copilot Studio), Neo4j GenAI, Local LLMs with Ollama

DevOps: Docker, Kubernetes, Git, CI/CD Pipelines

Testing: Validating query correctness and pipeline behavior against intended business logic, regression and release checks

Messaging & Monitoring: Kafka, RabbitMQ, Grafana, Log Analysis

Domain Exposure: Warehouse Automation, WMS Integration, Inventory Reconciliation, Robotics Telemetry, Computer Vision Data Integration

PROFESSIONAL EXPERIENCE

Software Engineer

Vimaan Robotics India Pvt Ltd | Bengaluru, India | Aug 2025 - Present

- Configure and optimize facility-specific Cypher and SQL queries that translate CV scan payloads — persisted through the Java pipeline — into structured inventory data surfaced directly on client-facing UI dashboards.
- Debug Cypher query failures and data pipeline issues across graph data flows, tracing root causes through query logic, relationship traversal, and backend-to-database integration paths.
- Built a Python-based warehouse configuration builder with a supporting UI front-end that semi-automated new warehouse onboarding, reducing the mapping effort from 2–3 days to under 1 day.
- Implement reconciliation logic that compares captured warehouse data against source records and surfaces mismatches, missing entries, and exception cases through Cypher queries and backend service checks.
- Debug Java/Spring Boot backend services and UI components rendering query outputs, validating dashboard behavior, API responses, and client-facing views across search, verification, and inventory workflows.
- Validate end-to-end pipeline behavior by tracing payloads through graph and backend layers, verifying query outputs and pipeline configurations against customer-defined business logic and expected data states.
- Support release validation and production debugging through log analysis, Cypher query inspection, and regression testing across backend services and database layers.

Senior Systems/Software Engineer

Cognizant Technology Solutions | Bengaluru, India | Sep 2022 - Jan 2025

- Developed and deployed Python/FastAPI and Node.js microservices using Docker and Kubernetes across AWS ECS and Azure AKS environments.
- Optimized PostgreSQL queries and data access patterns, reducing execution time by 25% and improving overall query performance by 19%.
- Built RESTful and GraphQL APIs with rate limiting, caching, OAuth 2.0, and JWT-based authentication, helping reduce API security risks.
- Developed event-driven workflows using Kafka and RabbitMQ, improving real-time processing efficiency while maintaining data consistency across services.
- Set up observability with Prometheus, Grafana, and ELK Stack to improve monitoring, debugging, and incident response workflows.
- Conducted code reviews and mentored junior engineers to improve code quality and delivery consistency.

PROJECTS

Olist Query Studio: Natural-Language to Cypher Workbench

Python, Neo4j, Cypher, Ollama, FastAPI, Streamlit, pytest | 2026

- Built a local workbench that turns plain-English analytics questions about the Olist e-commerce graph into parameterized Cypher via a local Ollama model grounded in the schema, with one automatic correction attempt on failure.
- Enforced read-only safety in layers: a query validator, an EXPLAIN check that rejects unknown labels, relationship types, and properties, read-mode transactions with timeouts and row limits, and a least-privilege, read-only Neo4j account.
- Built a 112-question verified benchmark and evaluation endpoint; the model currently matches 82 (73%), and integration tests re-run every reference query to catch data drift.

Bug2JSON: Bug Report Triage with LoRA Fine-Tuning

Python, PyTorch, Transformers, PEFT/LoRA, FastAPI, React, TypeScript, pytest | 2026

- Built a FastAPI service and React reviewer UI that turn free-text bug reports into schema-validated JSON triage records: category, severity, priority, environment, and probable cause.
- Implemented LoRA fine-tuning of Qwen2.5-1.5B-Instruct with prompt-loss masking, served beside the base model for comparison; data prep validates labels and keeps each issue group in one split to prevent leakage.
- Built a 30-case benchmark harness; the untuned base model scores 100% schema validity and 0.34 category macro-F1, the baseline for the fine-tune.

AI Resume Analyzer Web App

Next.js, React, TypeScript, Express.js, Supabase, OpenAI API, Razorpay, Zod | Ongoing

- Building a full-stack web app that uploads PDF/DOCX resumes, compares them with job descriptions or job links, and returns ATS-style feedback, keyword gaps, and improvement suggestions.
- Implemented a monorepo structure with a Next.js frontend, Express/TypeScript API, shared Zod schemas, Supabase authentication, protected routes, and database-backed analysis history.
- Developed resume analysis and editing workflows with OpenAI-assisted feedback, deterministic scoring logic, heuristic fallback behavior, structured resume parsing, diff output, and version history.
- Added supporting product features including public share reports, analytics events, background job handling, and Razorpay-based subscription flow experiments.
- Leveraged AI tooling for scaffolding and rapid UI iteration; personally designed the data flow, integrated all features, and validated end-to-end behavior.

Ethereal Command System Telemetry Dashboard

React, TypeScript, Vite, Tailwind CSS, Express.js, Recharts, systeminformation | Ongoing

- Built a local system monitoring dashboard with an Express backend that polls CPU, RAM, GPU, storage, thermal, battery, process, network interface, and socket telemetry.
- Designed cached local API endpoints and a React/Vite dashboard with live cards, time-series charts, resource views, process tables, network panels, and storage visualizations.
- Added process termination through a backend API with confirmation handling and error feedback for failed termination attempts.
- Leveraged AI tooling for component scaffolding and UI design passes; owned the backend architecture, API integration, and all feature validation.

EDUCATION

Master of Science in Information Technology

Ravenshaw University | Cuttack, India | Aug 2023 - Jul 2025

CGPA: 8.20 | Gold Medalist

Bachelor of Science in Information Technology and Management

Ravenshaw University | Cuttack, India | Aug 2019 - Jun 2022

CGPA: 8.68

CERTIFICATIONS

- **Neo4j GraphAcademy:** Neo4j Certified Professional, 2026; Neo4j Graph Data Science, 2026; Neo4j & Generative AI, 2026
- **Microsoft:** Applied Skills: Get started developing agents in Microsoft Foundry, 2026; Enhance agents with autonomous capabilities, 2026
- **HackerRank:** SQL Advanced, 2023; Software Engineer, 2025; Python, 2025; Frontend Developer React, 2025
- **Udemy:** Modern JavaScript, 2024; Advanced SQL: MySQL Data Analysis and Business Intelligence, 2023

HONORS & AWARDS

- Gold Medalist, Master of Science in Information Technology, Ravenshaw University; medal presented by the President of India at the university convocation.
- Cognizant Excellence Award for measurable improvements in database query performance and system efficiency, enhancing client satisfaction.
- Grade A recognition for final-year research on advanced data security methodologies.